

MANAM THULASIRAM

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CAREER OBJECTIVE

Final-year B.Tech student in Computer Science and Engineering (Data Science) with practical experience in machine learning, full-stack development, and data-driven problem solving. Seeking an entry-level software or data engineering role where strong programming fundamentals, end-to-end project delivery, and a peer-reviewed research publication contribute to team goals from day one.

EDUCATION

B.Tech – Computer Science and Engineering (Data Science) Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai	Expected: 2027
Diploma – Computer Engineering Sri Venkatesa Perumal College of Engineering and Technology, Puttur	Completed: 2024
Secondary School Certificate (SSC) Z.P. High School, Arai, Tirupathi District, Andhra Pradesh	Completed: 2021

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML, CSS
ML & AI: YOLOv8, PyTorch, Scikit-Learn, NumPy, Pandas, TensorFlow, XGBoost, SHAP, Autoencoders
Backend & APIs: FastAPI, RESTful API Design, JWT Authentication, bcrypt
Frontend & Mobile: React, React Native, TypeScript, Expo, Ngix
Databases: MySQL, MongoDB, PostgreSQL, SQLite
Data & Analytics: Excel, Power BI, Jupyter Notebook, Google Colab, Kaggle
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

NutriVision – AI-Powered Nutrition Intelligence & Recommendation Application

Tech Stack: *Python, FastAPI, YOLOv8, PyTorch, Scikit-Learn (KNN), NumPy, Pandas, React Native, TypeScript, Expo, PostgreSQL, SQLite, Docker, Nginx, GitHub Actions, Prometheus, Sentry, Pytest, Locust, JWT, bcrypt*

- Developed a full-stack mobile application that automates food recognition, nutrient analysis, and personalized meal recommendations using computer vision and machine learning.
- Integrated YOLOv8 for regional food detection and PyTorch Autoencoders with KNN-based filtering to deliver intelligent health recommendations in real time.
- Built a RESTful backend with JWT authentication and bcrypt; deployed via Docker and Nginx with CI/CD through GitHub Actions.
- Validated system reliability using Pytest and Locust load testing; monitored application performance with Prometheus and Sentry.
- Packaged the application as an Android APK; trained and validated models using structured Kaggle nutritional datasets.

RESEARCH & PUBLICATION

Presented and published “Precision Dietary Recommendation using YOLOv8 Food Detection and Optimization-based Nutritional Planning” at the International Conference on Innovations in Cybersecurity and Data Science (ICICDS 2026).

CERTIFICATIONS

- AWS Certified Machine Learning
- Big Data Certification
- Data Structures & Algorithms in Python: Sorting Algorithms
- NPTEL – Human Computer Interaction (**Elite – 90%**)
- NPTEL – Introduction to Industry 4.0 & Industrial Internet of Things
- NPTEL – Remote Sensing Essentials

WORKSHOPS & EVENTS

- Applied ML: Where Ideas Meet Reality – Workshop
- Codeathon 4.0 – Competitive Programming Participant
- BIOSPARX 2K26 – Technical Event Participant
- EDZIAL 2.0 – Technical Fest Participant
- TECHYUGAM '26 – Technical Symposium Participant

CORE COMPETENCIES

Problem Solving | Analytical Thinking | Team Collaboration | Written & Verbal Communication | Time Management | Continuous Learning